FOR THE STUDENT ASSISTANT · SYMPTOM → CAUSE → FIX

Colab troubleshooting

NB0 setup · NB1–2 GPT-2, CPU ok · NB3 T4 (GPT-2 on CPU)
NB4 fresh runtime, Neuronpedia path · NB5 cache, CPU

First 60 seconds with any problem

1. Read the **last line** of the red error. Find it in the table below.
2. Is the participant on **their own copy**? (**File › Save a copy in Drive**.)
3. Check runtime type: **Runtime › Change runtime type**. NB1–2 and NB5 run on CPU. NB4 always in a **fresh runtime**.
4. Still stuck after 3 minutes: **pair them with a neighbour** and follow along on one screen. Come back at the break.

Fallbacks that always work

- **No GPU:** NB1, NB2, NB5 run on CPU. NB3 falls back to GPT-2 automatically. NB4's default path is Neuronpedia (no model).
- **No Hugging Face access:** watch the projector demo; open the Neuronpedia graph in a browser.
- **Laptop dead:** Colab works in a phone browser for read-along.

Symptom (what they see)	Likely cause	Fix
RUNTIME AND GPU		
"Cannot connect to GPU backend" / no T4 offered `GPU usage limit reached`	Free-tier GPU quota used up, or no GPUs free in the region right now.	Switch to CPU and continue. NB1, NB2 and NB5 are CPU-ready. NB3 switches to GPT-2 by itself. NB4 falls back to `PATH = "neuronpedia"`. Try a different Google account if they have one. Do not wait in a retry loop.
How do I get a GPU?	Runtime is CPU by default.	**Runtime › Change runtime type › Hardware accelerator: T4 GPU › Save**. This restarts the runtime: re-run the install and setup cells. Check with `!nvidia-smi`.
"Runtime disconnected" / cells stop with no output	Idle timeout, laptop slept, Wi-Fi dropped, or the session crashed (often RAM).	Click **Reconnect**, then **Runtime › Run before** on the current cell. Variables are lost on disconnect; files in `/content` may also be gone. Keep the laptop plugged in and the tab visible.
"Your session crashed after using all available RAM"	System RAM full (12 GB): model loaded twice, or caching all activations.	**Runtime › Restart session**, run cells once from the top. Do not re-run the model-loading cell repeatedly.
INSTALLS AND IMPORTS		
pip shows red "dependency conflicts" text `ERROR: pip's dependency resolver ...`	Colab's preinstalled packages disagree with our pinned versions. Usually harmless.	If the cell finishes, ignore the warning. If Colab shows a **Restart session** button, click it, then run from the cell after the install.
`ImportError` / `AttributeError` right after installing	Old version still loaded in memory (e.g. numpy, transformers).	**Runtime › Restart session** (not "disconnect and delete"), then continue below the install cell. Do not re-run pip.
NB4: `ModuleNotFoundError: nnsight.intervention.tracing`, or version errors after the install cell	NB4 needs its own older stack; nnsight 0.8.x breaks circuit-tracer 0.5.0. Runtime still holds NB0–3 packages.	**Runtime › Disconnect and delete runtime**, reconnect, and run NB4 from the top in this **fresh runtime**. The pins are `circuit-tracer==0.5.0 nnsight==0.7.0 transformers==4.57.3 huggingface_hub==0.36.2 transformer_lens==3.2.1`. Never run NB4 after NB3 in the same runtime.
`ModuleNotFoundError: transformer_lens` (or sae_lens)	Install cell skipped, or runtime was reset (e.g. after changing the runtime type).	Run the install cell again, wait for it to finish, then the setup cell.
HUGGING FACE AND GEMMA ACCESS		
"Secret HF_TOKEN does not exist" / login prompt / `401`	No token in Colab, notebook access not granted, or token mistyped / revoked.	Get a token at `huggingface.co/settings/tokens` (type **Read**). In Colab: key icon on the left › **Add new secret** › name `HF_TOKEN`, paste value › turn on **Notebook access**. Re-run the setup cell.
`403` / `GatedRepoError` / "Access to model google/gemma-... is restricted"	Gemma license not accepted on this Hugging Face account.	Open the model page while logged in and click **Acknowledge license**: `google/gemma-3-1b-it` (NB3), `google/gemma-3-4b-it` (NB5 live, optional), `google/gemma-2-2b` (NB4 opt-in). Approval is usually instant; re-run the cell. Token must belong to the same account. NB0 checks all three.
MEMORY, SPEED, NETWORK		
`CUDA out of memory` on T4	15 GB GPU shared by model, SAE and caches; or a previous model still loaded.	Restart session and run once from the top. NB3 must stay in **float32** (T4 has no bf16; Gemma overflows fp16, giving NaNs/garbage); it needs only ~4 GB. NB4: use `PATH = "neuronpedia"`; the Gemma 2 2B path is opt-in. Use `names_filter`; lower batch size.
Model or SAE download crawls / times out	20 laptops on one Wi-Fi pulling GBs at once.	NB3 first run downloads ~2 GB: start it at the break. NB4's Gemma 2 2B path is ~18 GB; do not enable it in the room. NB5 needs only its small cache. If the venue Wi-Fi fails, tether to a phone hotspot for small files only.
Neuronpedia page blank or very slow	Site load, browser extensions, or blocked third-party scripts.	Reload; try a private window or another browser; disable ad-blockers for neuronpedia.org. Otherwise follow the projector.
NOTEBOOK HANDLING		
"You are viewing a read-only notebook" / changes not saved	Opened from the GitHub link without copying.	**File › Save a copy in Drive**. Work in the copy (title starts with "Copy of").
Participant wants to share their notebook with a partner	—	**Share** (top right) › add the partner's email as Editor, or **General access: Anyone with the link** › Copy link. Each person runs their own runtime; code is shared, variables are not.

Escalate to the presenter only if 3+ people hit the same error.